

MTH 166

Lecture-32

Divergence and Curl of a Vector Field

Topic:

Vector Differential Calculus

Learning Outcomes:

1. To calculate divergence of a vector field
2. To calculate curl of a vector field

Let the given vector field be $\vec{v} = v_1\hat{i} + v_2\hat{j} + v_3\hat{k}$

Let us consider a vector differential operator $\vec{\nabla} = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right)$

Divergence: It is dot product between $\vec{\nabla}$ and $\vec{v}$ and is given by:

$$\begin{aligned} \text{div}(\vec{v}) &= \vec{\nabla} \cdot \vec{v} = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot (v_1\hat{i} + v_2\hat{j} + v_3\hat{k}) \\ &= \frac{\partial v_1}{\partial x} + \frac{\partial v_2}{\partial y} + \frac{\partial v_3}{\partial z} \quad (\because \hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1) \end{aligned}$$

Divergence is always a scalar quantity (Having no direction).

Note: If $\text{div}(\vec{v}) = 0$, then vector $\vec{v}$ is called **Solenoidal** or **Incompressible**.

That means the field has no sources and no sinks.

Curl: It is cross product between $\vec{\nabla}$ and $\vec{v}$ and is given by:

$$\text{curl}(\vec{v}) = \vec{\nabla} \times \vec{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ v_1 & v_2 & v_3 \end{vmatrix}$$

Curl is always a vector quantity.

Note: If $\text{curl}(\vec{v}) = \vec{0}$, then vector $\vec{v}$ is called **Irrotational** or **Conservative**.

Two Important Properties:

1. $\text{div}(\text{curl}(\vec{v})) = 0$ (Always)
2. $\text{curl}(\text{grad}(f)) = \vec{0}$ (Always)

Problem 1. Compute $\text{div}(\vec{v})$ and $\text{curl}(\vec{v})$ and verify that $\text{div}(\text{curl}(\vec{v})) = 0$ where

$$\vec{v} = x\hat{i} + 2y\hat{j} + z\hat{k}$$

Solution. Here $\vec{v} = x\hat{i} + 2y\hat{j} + z\hat{k}$

Comparing with: $\vec{v} = v_1\hat{i} + v_2\hat{j} + v_3\hat{k}$

$$v_1 = x, \quad v_2 = 2y, \quad v_3 = z$$

$$\Rightarrow \frac{\partial v_1}{\partial x} = 1, \quad \frac{\partial v_2}{\partial y} = 2, \quad \frac{\partial v_3}{\partial z} = 1$$

$$\text{div}(\vec{v}) = \vec{\nabla} \cdot \vec{v} = \frac{\partial v_1}{\partial x} + \frac{\partial v_2}{\partial y} + \frac{\partial v_3}{\partial z}$$

$$= 1 + 2 + 1 = 4$$

$$\begin{aligned}
\text{curl}(\vec{v}) &= \vec{\nabla} \times \vec{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ v_1 & v_2 & v_3 \end{vmatrix} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x & 2y & z \end{vmatrix} \\
&= \hat{i} \left(\frac{\partial(z)}{\partial y} - \frac{\partial(2y)}{\partial z} \right) - \hat{j} \left(\frac{\partial(z)}{\partial x} - \frac{\partial(x)}{\partial z} \right) + \hat{k} \left(\frac{\partial(2y)}{\partial x} - \frac{\partial(x)}{\partial y} \right) \\
&= 0\hat{i} - 0\hat{j} + 0\hat{k} = \vec{0} \text{ (Irrotational)}
\end{aligned}$$

$$\text{div}(\text{curl}(\vec{v})) = \text{div}(\vec{0}) = 0$$

Problem 2. Compute $\text{div}(\vec{v})$ and $\text{curl}(\vec{v})$ and verify that $\text{div}(\text{curl}(\vec{v})) = 0$ where

$$\vec{v} = xy\hat{i} + yz\hat{j} + zx\hat{k}$$

Solution. Here $\vec{v} = xy\hat{i} + yz\hat{j} + zx\hat{k}$

Comparing with: $\vec{v} = v_1\hat{i} + v_2\hat{j} + v_3\hat{k}$

$$v_1 = xy, \quad v_2 = yz, \quad v_3 = zx$$

$$\Rightarrow \frac{\partial v_1}{\partial x} = y, \quad \frac{\partial v_2}{\partial y} = z, \quad \frac{\partial v_3}{\partial z} = x$$

$$\text{div}(\vec{v}) = \vec{\nabla} \cdot \vec{v} = \frac{\partial v_1}{\partial x} + \frac{\partial v_2}{\partial y} + \frac{\partial v_3}{\partial z}$$

$$= y + z + x$$

$$\begin{aligned}
\text{curl}(\vec{v}) &= \vec{\nabla} \times \vec{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ v_1 & v_2 & v_3 \end{vmatrix} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xy & yz & zx \end{vmatrix} \\
&= \hat{i} \left(\frac{\partial(zx)}{\partial y} - \frac{\partial(yz)}{\partial z} \right) - \hat{j} \left(\frac{\partial(zx)}{\partial x} - \frac{\partial(xy)}{\partial z} \right) + \hat{k} \left(\frac{\partial(yz)}{\partial x} - \frac{\partial(xy)}{\partial y} \right) \\
&= -y\hat{i} - z\hat{j} - x\hat{k}
\end{aligned}$$

$$\begin{aligned}
\text{div}(\text{curl}(\vec{v})) &= \text{div}(-y\hat{i} - z\hat{j} - x\hat{k}) = \frac{\partial(-y)}{\partial x} + \frac{\partial(-z)}{\partial y} + \frac{\partial(-x)}{\partial z} \\
&= 0 + 0 + 0 = 0
\end{aligned}$$

Polling quiz

Let $\vec{v}$ be a differentiable vector field. Then:

(a) $\operatorname{div}(\operatorname{curl} \vec{v}) = \vec{v}$

(b) $\operatorname{div}(\operatorname{curl} \vec{v}) \neq 0$

(c) $\operatorname{div}(\operatorname{curl} \vec{v}) = 0$

(d) $\operatorname{div}(\operatorname{curl} \vec{v}) = \hat{i} + \hat{j} + \hat{k}$

Problem 3. Compute $\text{grad}(f)$ and verify that $\text{curl}(\text{grad}(f)) = \vec{0}$ where $f = x + y - 2z^2$

Solution. Let $f(x, y, z) = x + y - 2z^2$

$$\Rightarrow f_x = 1, f_y = 1, f_z = -4z$$

Then gradient of scalar field f is given by:

$$\text{grad}(f) = \vec{\nabla} f = (f_x \hat{i} + f_y \hat{j} + f_z \hat{k}) = \hat{i} + \hat{j} - 4z \hat{k}$$

$$\text{curl}(\text{grad}(f)) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 1 & 1 & -4z \end{vmatrix} = 0\hat{i} + 0\hat{j} + 0\hat{k} = \vec{0}$$

Problem 4. Compute $\text{grad}(f)$ and verify that $\text{curl}(\text{grad}(f)) = \vec{0}$ where $f = 16xy^3z^2$

Solution. Let $f(x, y, z) = 16xy^3z^2$

$$\Rightarrow f_x = 16y^3z^2, f_y = 48xy^2z^2, f_z = 32xy^3z$$

Then gradient of scalar field f is given by:

$$\text{grad}(f) = \vec{\nabla}f = (f_x\hat{i} + f_y\hat{j} + f_z\hat{k}) = 16y^3z^2\hat{i} + 48xy^2z^2\hat{j} + 32xy^3z\hat{k}$$

$$\text{curl}(\text{grad}(f)) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 16y^3z^2 & 48xy^2z^2 & 32xy^3z \end{vmatrix} = 0\hat{i} + 0\hat{j} + 0\hat{k} = \vec{0}$$

Polling quiz

Let f be a differentiable scalar field. Then:

(a) $\operatorname{curl}(\operatorname{grad} f) = \vec{0}$

(b) $\operatorname{curl}(\operatorname{grad} f) \neq \vec{0}$

(c) $\operatorname{curl}(\operatorname{grad} f) = \vec{f}$

(d) $\operatorname{curl}(\operatorname{grad} f) = \hat{i} + \hat{j} + \hat{k}$

1. If $\text{div}(\vec{v}) = 0$, then vector $\vec{v}$ is called **Solenoidal** or **Incompressible**.

That means the field has no sources and no sinks.

2. If $\text{curl}(\vec{v}) = \vec{0}$, then vector $\vec{v}$ is called **Irrotational** or **Conservative**.

Problem 1. Show that the vector $\vec{v} = (2x + 3y)\hat{i} + (x - y)\hat{j} - (x + y + z)\hat{k}$ is solenoidal.

Solution. Here $\vec{v} = (2x + 3y)\hat{i} + (x - y)\hat{j} - (x + y + z)\hat{k}$

Comparing with: $\vec{v} = v_1\hat{i} + v_2\hat{j} + v_3\hat{k}$

$$v_1 = (2x + 3y), \quad v_2 = (x - y), \quad v_3 = -(x + y + z)$$

$$\Rightarrow \frac{\partial v_1}{\partial x} = 2, \quad \frac{\partial v_2}{\partial y} = -1, \quad \frac{\partial v_3}{\partial z} = -1$$

$$\begin{aligned} \text{div}(\vec{v}) &= \vec{\nabla} \cdot \vec{v} = \frac{\partial v_1}{\partial x} + \frac{\partial v_2}{\partial y} + \frac{\partial v_3}{\partial z} \\ &= 2 - 1 - 1 = 0 \end{aligned}$$

So, the given vector $\vec{v}$ is Solenoidal.

Problem 2. Show that the vector $\vec{v} = 3x^2y^2z^4\hat{i} + 2x^3yz^4\hat{j} + 4x^3y^2z^3\hat{k}$ is Irrotational.

Solution. Here $\vec{v} = 3x^2y^2z^4\hat{i} + 2x^3yz^4\hat{j} + 4x^3y^2z^3\hat{k}$

Comparing with: $\vec{v} = v_1\hat{i} + v_2\hat{j} + v_3\hat{k}$

$$\begin{aligned}\text{curl}(\vec{v}) &= \vec{\nabla} \times \vec{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ v_1 & v_2 & v_3 \end{vmatrix} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 3x^2y^2z^4 & 2x^3yz^4 & 4x^3y^2z^3 \end{vmatrix} \\ &= \hat{i}(8x^3yz^3 - 8x^3yz^3) - \hat{j}(12x^2y^2z^3 - 12x^2y^2z^3) + \hat{k}(6x^2yz^4 - 6x^2yz^4) \\ &= 0\hat{i} - 0\hat{j} + 0\hat{k} = \vec{0}\end{aligned}$$

So, the given vector $\vec{v}$ is Irrotational.

